

Cell 1: Project Title (Markdown)

Marketing Funnel & Conversion Performance Analysis

FUTURE_DS_03 – Future Interns Internship

This notebook analyzes marketing funnel data to identify conversion drop-offs, channel performance, and opportunities to improve lead-to-customer conversion rates.

Cell 2: Import Libraries (Code)

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

Cell 3: Load Dataset (Code)

```
df = pd.read_csv("marketing_funnel_data.csv")

df.head()
```

Cell 4: Dataset Information (Code)

```
print("Dataset Shape:", df.shape)

df.info()

df.describe()
```

Cell 5: Check Missing Values (Code)

```
print(df.isnull().sum())
```

Cell 6: Data Cleaning (Code)

```
df.drop_duplicates(inplace=True)

print("Updated Shape:", df.shape)
```

Cell 7: Funnel Metrics Calculation (Code)

```
total_visitors = df['Visitors'].sum() total_leads = df['Leads'].sum() total_qualified = df['Qualified_Leads'].sum()
total_customers = df['Customers'].sum()

visitor_to_lead = (total_leads / total_visitors) * 100 lead_to_qualified = (total_qualified / total_leads) * 100
qualified_to_customer = (total_customers / total_qualified) * 100 overall_conversion = (total_customers /
total_visitors) * 100

print("Visitor to Lead Conversion:", round(visitor_to_lead,2), "%") print("Lead to Qualified Lead Conversion:",
round(lead_to_qualified,2), "%") print("Qualified Lead to Customer Conversion:",
round(qualified_to_customer,2), "%") print("Overall Conversion Rate:", round(overall_conversion,2), "%")
```

Cell 8: Funnel Visualization (Code)

```
funnel_data = [total_visitors, total_leads, total_qualified, total_customers]

stages = ['Visitors', 'Leads', 'Qualified Leads', 'Customers']

plt.figure(figsize=(8,5)) sns.barplot(x=stages, y=funnel_data) plt.title("Marketing Funnel") plt.ylabel("Count")
plt.show()
```

Cell 9: Channel Performance Analysis (Code)

```
channel_summary = df.groupby('Channel')['Customers'].sum().sort_values(ascending=False)

print(channel_summary)
```

Cell 10: Channel Performance Chart (Code)

```
plt.figure(figsize=(8,5)) channel_summary.plot(kind='bar') plt.title("Customers by Marketing Channel")  
plt.ylabel("Customers") plt.show()
```

Cell 11: Conversion Rate by Channel (Code)

```
df['Channel_Conversion_Rate'] = ( df['Customers'] / df['Visitors'] ) * 100  
  
channel_conversion = df.groupby('Channel')['Channel_Conversion_Rate'].mean()  
  
print(channel_conversion)
```

Cell 12: Conversion Rate Visualization (Code)

```
plt.figure(figsize=(8,5)) channel_conversion.plot(kind='bar') plt.title("Average Conversion Rate by Channel")  
plt.ylabel("Conversion Rate (%)") plt.show()
```

Cell 13: Drop-Off Analysis (Code)

```
drop_off_1 = total_visitors - total_leads drop_off_2 = total_leads - total_qualified drop_off_3 = total_qualified -  
total_customers  
  
drop_offs = [drop_off_1, drop_off_2, drop_off_3]  
  
stages = [ "Visitors → Leads", "Leads → Qualified", "Qualified → Customers" ]  
  
plt.figure(figsize=(8,5)) sns.barplot(x=stages, y=drop_offs) plt.title("Funnel Drop-Off Analysis")  
plt.ylabel("Drop-Off Count") plt.show()
```

Cell 14: Key Insights (Markdown)

Key Insights

1. Identified major drop-off stages in the marketing funnel.
2. Measured overall conversion performance.
3. Evaluated channel effectiveness.

4. Compared conversion rates across channels.
 5. Generated recommendations for funnel optimization.
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Cell 15: Recommendations (Markdown)

Recommendations

- Improve landing page performance.
 - Optimize lead nurturing campaigns.
 - Focus investment on high-performing channels.
 - Conduct A/B testing.
 - Improve customer engagement strategies.
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Cell 16: Conclusion (Markdown)

Conclusion

This analysis successfully evaluated marketing funnel performance, identified conversion bottlenecks, and provided actionable recommendations to improve lead-to-customer conversion rates and overall marketing ROI.